

CIVL6415, Semester 2, 2025

TRAFFIC ANALYSIS AND SIMULATION

MODULE 5

Traffic Assignment

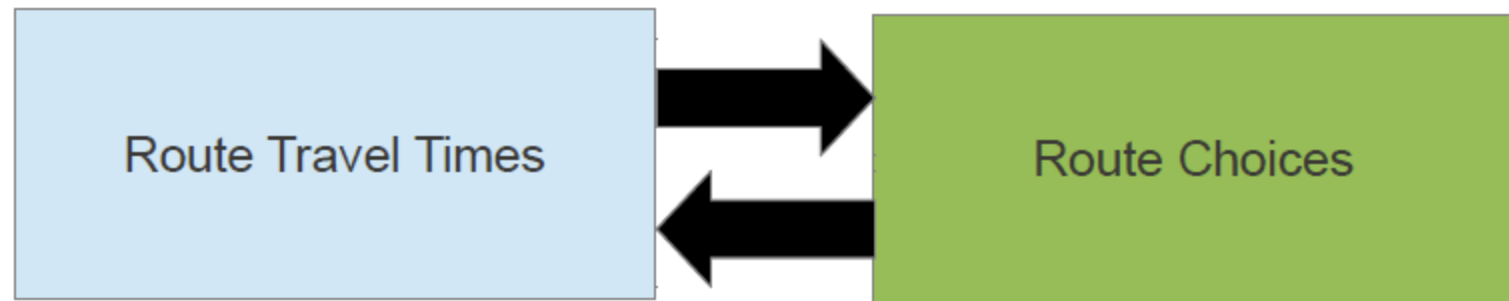
Lecture Week 11

Dr. Mehmet Yildirimoglu
School of Civil Engineering
The University of Queensland
m.yildirimoglu@uq.edu.au

- Introduction to traffic assignment / equilibrium
- Path choice models
 - Deterministic vs. Probabilistic
- Static traffic assignment
 - User Equilibrium (UE)
 - System Optimum (SO)
 - Methods to reach equilibrium
- Dynamic traffic assignment
 - Dynamic User Equilibrium (DUE), Dynamic System Optimum (DSO)
 - Method of Successive Averages (MSA)
- Departure time choice

- Transportation systems involve interactions among multiple agents. The basic facts are:
 - Although travel choices are made individually, they impact others as well (congestion).
 - The impacts of others' travel choices are important as you make your choices.

It's the evening peak period, so the freeway will probably be congested. I'll take another route." (But isn't everyone else thinking the same way?)

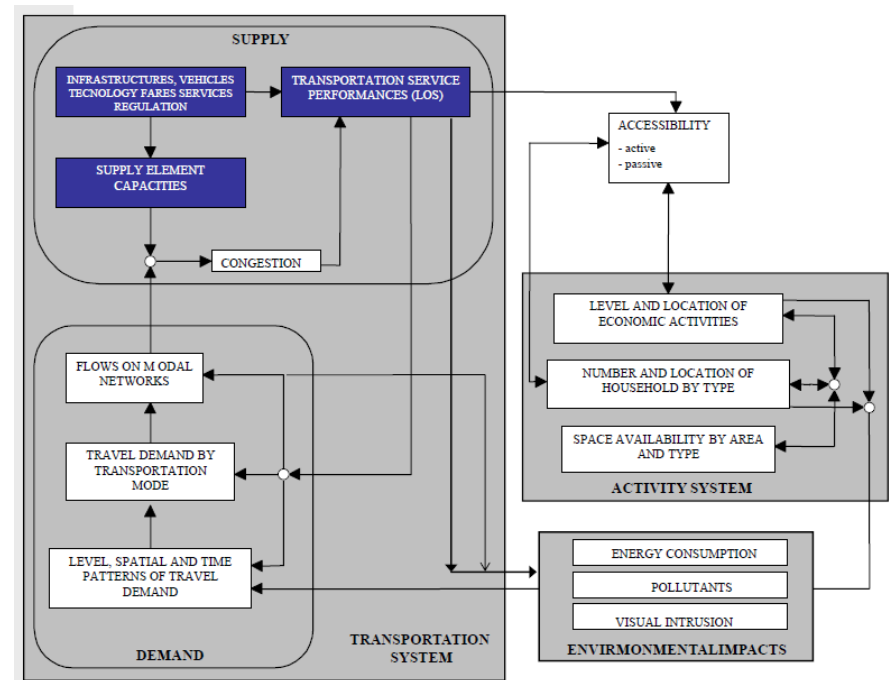


▪ **4-Step Modelling**

- Trip generation (Estimate trips produced and attracted in each zone)
- Trip distribution (Find O-D flows)
- Mode split (Consider alternative transport modes)
- **Traffic assignment (Route choice for O-D car flows)**
 - **Equilibrium**

Introduction

Simulate the way in which demand and supply interact in transportation systems, resulting in flows and performance levels on network elements.

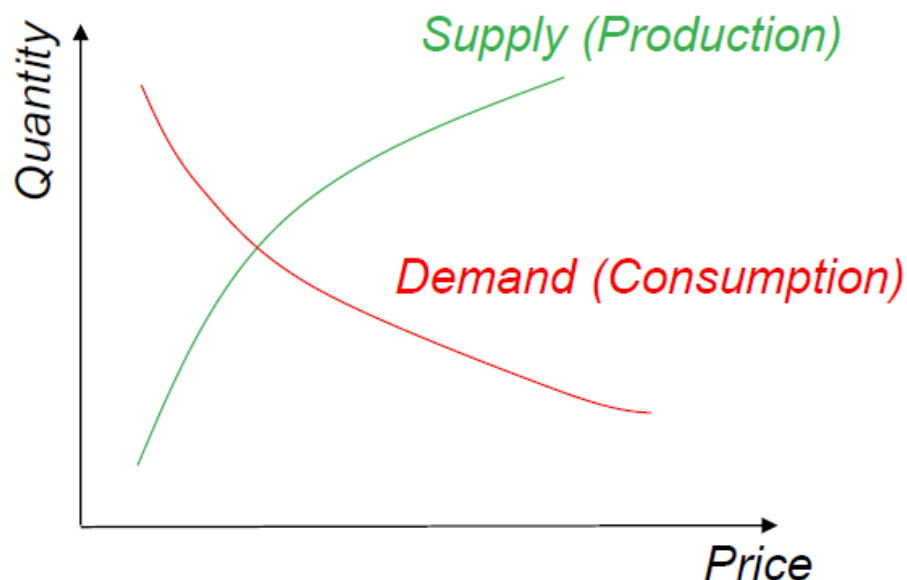


Supply / Demand Interactions

- **Demand curve** represents the amount of some goods that buyers are willing and able to purchase at various prices
- **Supply curve** represents the amount of some goods that producers are willing to provide at various prices

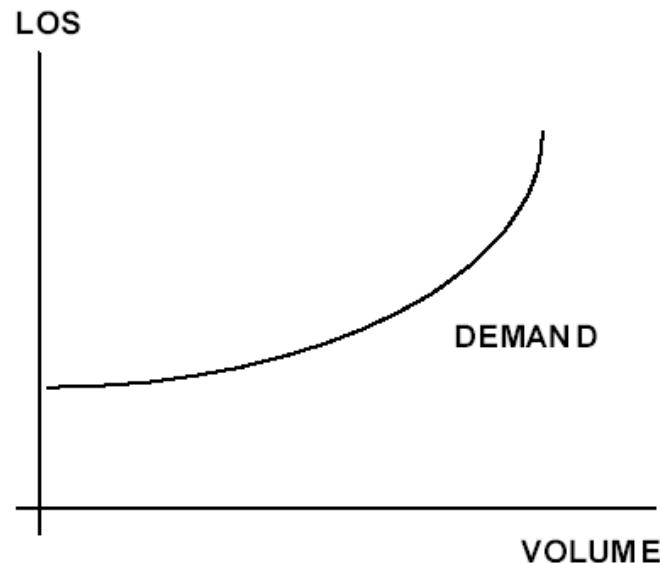
- This is simply a micro-economic concept. For example, if a movie theater doubles its price, therefore making its service less attractive -- in this case, more expensive -- fewer people will go to that movie theater. If a movie theater halves its price, more people will go.

In economics



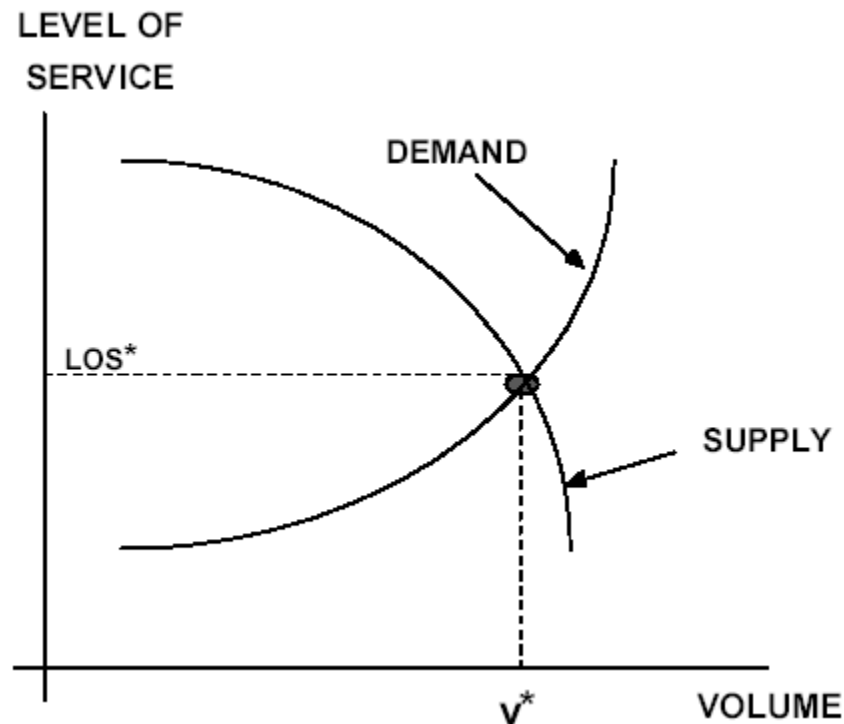
Demand curve

- The volume that a transportation service attracts is a **function of the level-of-service** provided to customers. If the level-of-service deteriorates, less people will want to use the service.
- LOS similar to utility, a measure of benefit or satisfaction. Might be linear sum of various LOS variables, such as travel time, access time, waiting time, fare and comfort. Inverse relationship between travel time and LOS



Supply curve

- Level-of-service as **a function of volume**.
- As a facility gets congested, the level-of-service for customers deteriorates
- **At equilibrium:** Quantity produced = Quantity demanded



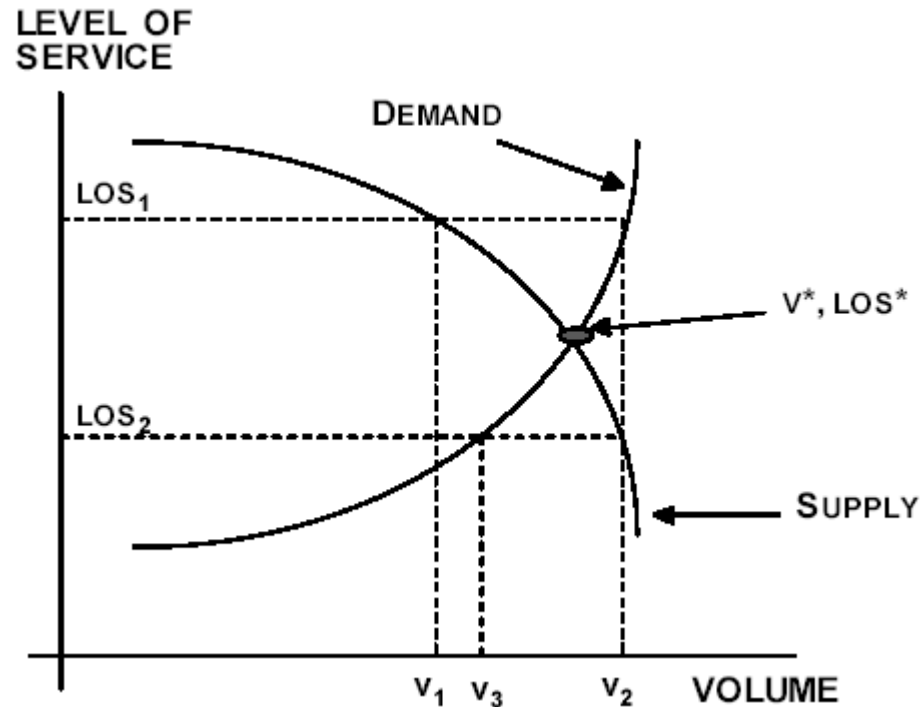
Iteration to find equilibrium

v_1 --first guess at volume

LOS_1 implies demand v_2

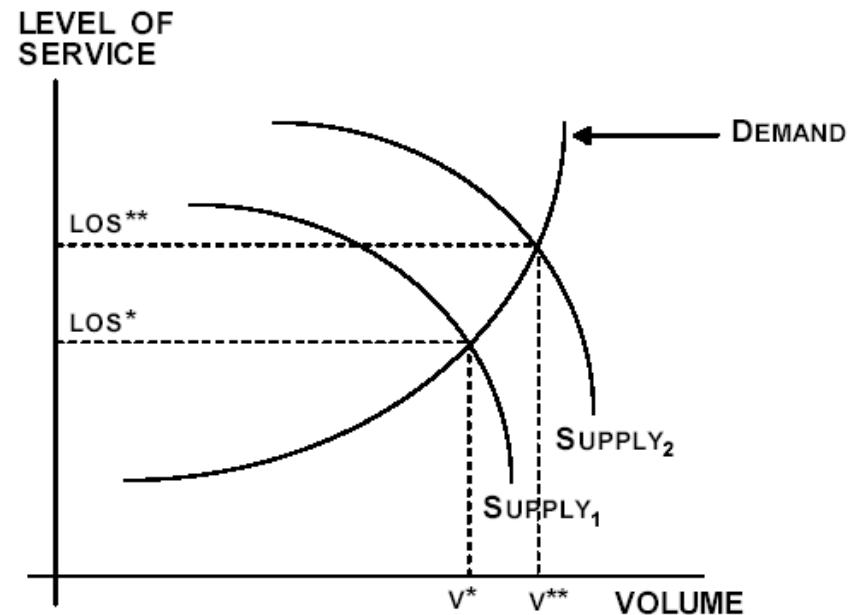
v_2 implies LOS_2

LOS_2 implies v_3 , etc., until v^* , LOS^* are found.



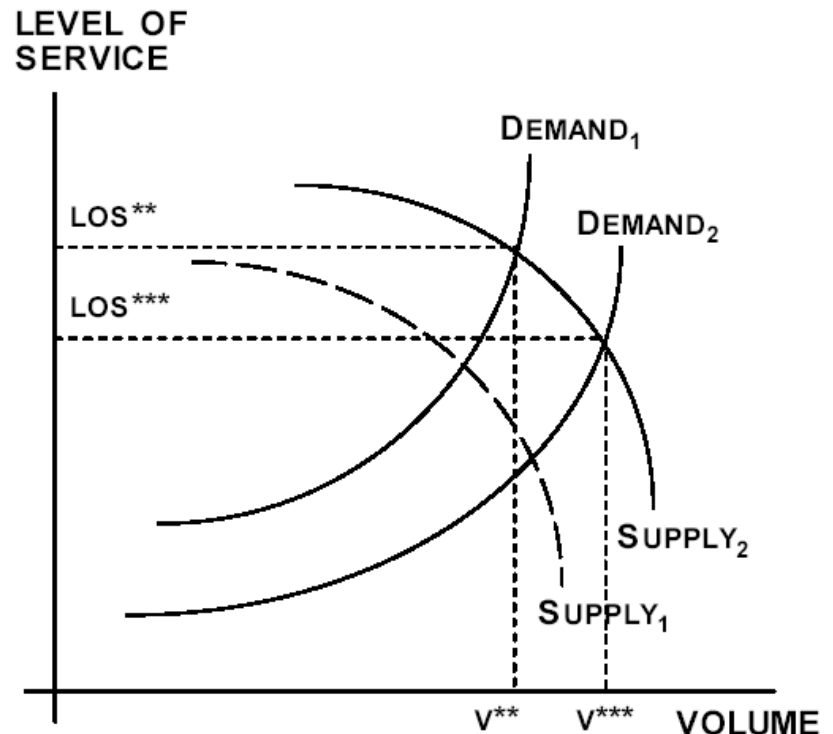
Measures to increase supply

- General street traffic engineering improvements
 - minor design improvements, improved
 - signal timing
- Freeway traffic management – ramp metering
- From Supply 1 to Supply 2

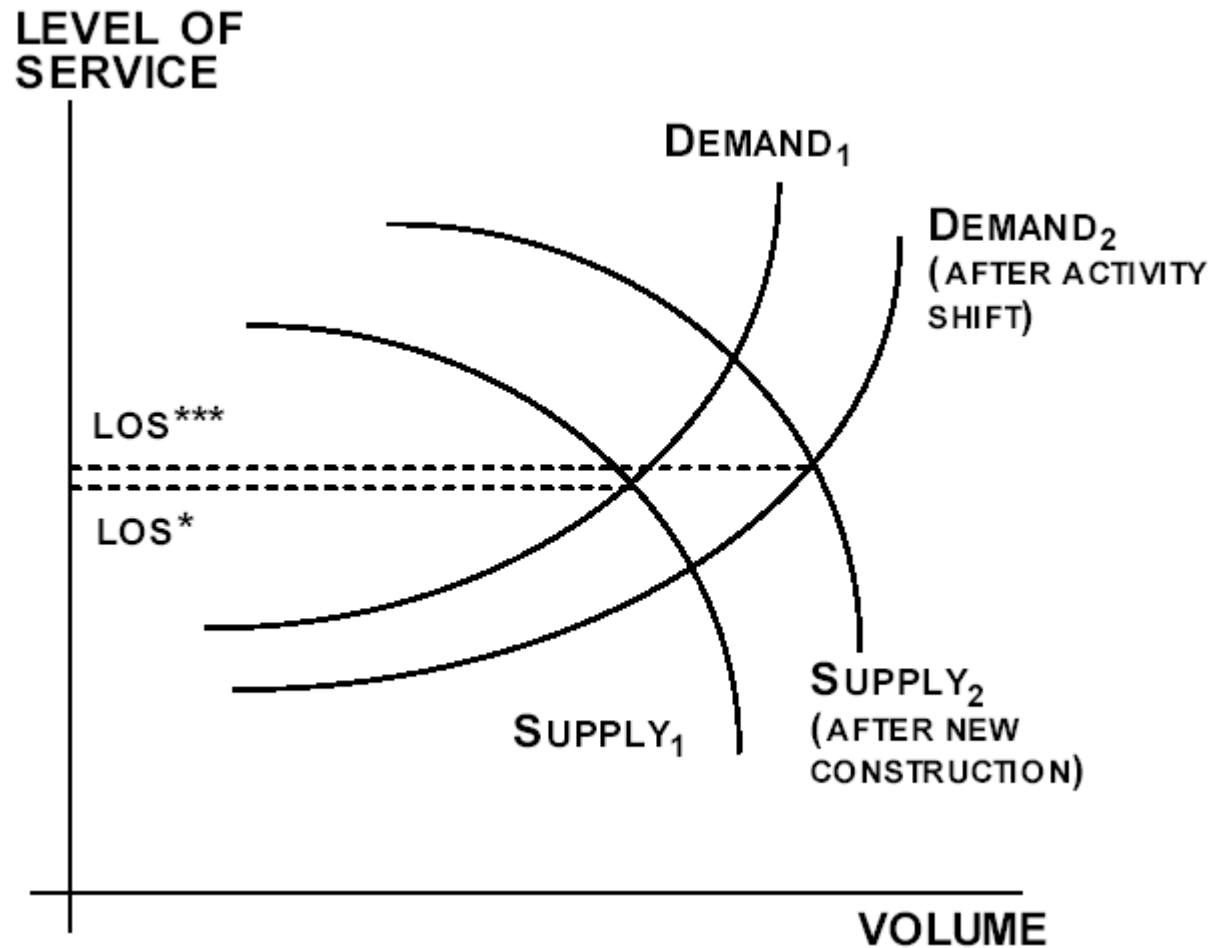


Measures to reduce demand

- Ridesharing, carpooling
- Park-and-ride
- Telecommuting
- Promoting off-peak travel
- From Demand 2 to Demand 1

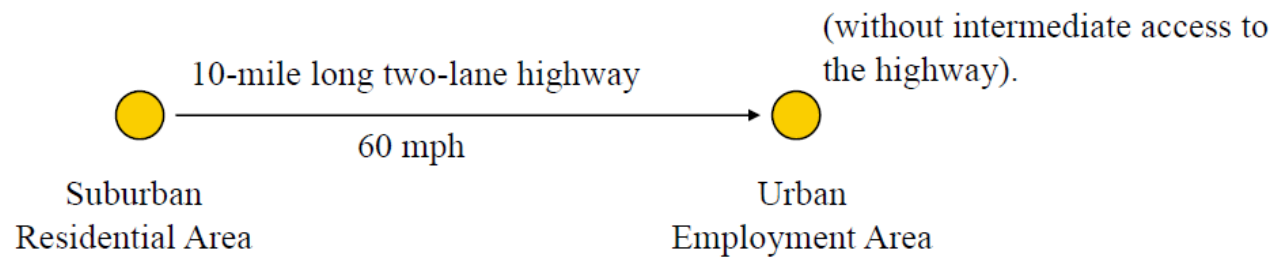


Induced demand



Exercise

Consider



Exercise

Assume a simple, linear model specification for Supply Function:

$$t = m + n \cdot V$$

where:

t = travel time (min.) a measure of LOS

V = volume (vehicle per hour, vph)

m, n = parameters defining the Transportation System ' T ' = $\{m, n\}$

Let $m = 10$ min. (corresponding to a mean speed of 60 mph over the 10 mile highway)

$n = 0.01$ min. per vph (the sensitivity of travel time to increases in volume)

Thus:

$$t = 10 + 0.01V$$

Exercise

Again, assume a simple, linear model specification for Demand Function:

$$V = a + b \cdot t$$

where:

a, b = parameters defining the Activity System ' A ' = {a,b}

Let a = 5000 vph (corresponding to a potential, or latent, demand for the 10 mile highway with zero travel time)

b = -100 vph per minute (the sensitivity of demand to increases in travel time)

Thus:

$$V = 5000 - 100t$$

Exercise

The Flow Pattern '**F**' defined by '**V**' and '**S**' is the resultant of the equilibration of the Service Function '**J**' and the Demand Function '**D**'.

For this simple example, '**F**' may be determined either graphically or by solving the two functions simultaneously.

Travel Market Equilibration

The equilibrium flow pattern is:

$$t = 10 + 0.01V$$

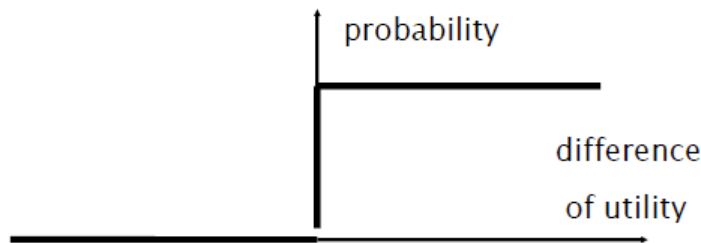
$$V = 5000 - 100t \Rightarrow F1 = \{2000 \text{ vph, } 30.0 \text{ minutes}\}$$



Path choice

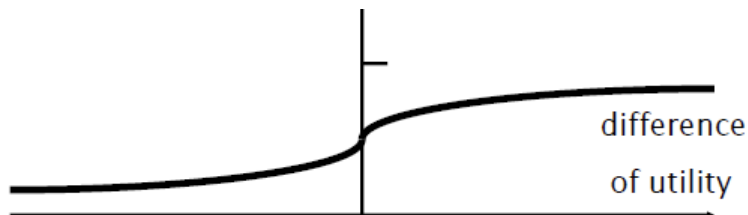
Deterministic

Users choose only the minimum cost alternative



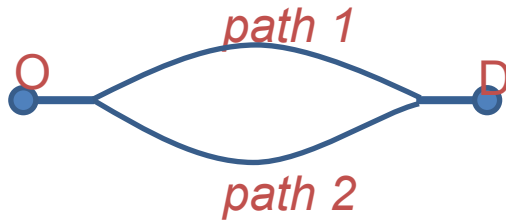
Probabilistic

Users may choose any available alternative with a probability depending on costs



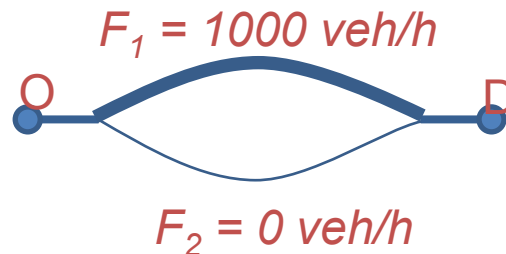
Example

Deterministic vs. Probabilistic path choice



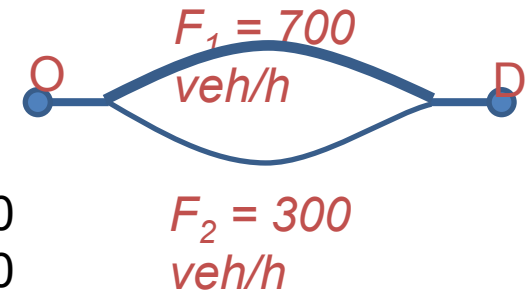
$d_{OD} = 1000 \text{ veh/h}$
Path 1: $t_1 = 10 \text{ min}$
Path 2: $t_2 = 15 \text{ min}$

Deterministic



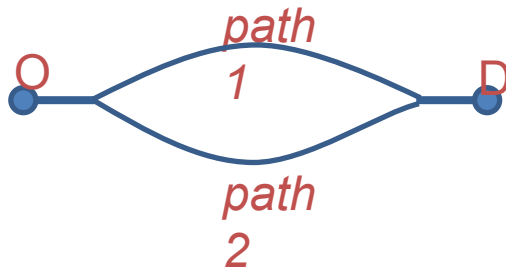
Probabilistic

$V_1 = \alpha \cdot t_1$
 $V_2 = \alpha \cdot t_2$
 $\alpha = -5.90$
Prob[1] = 0.70
Prob[2] = 0.30

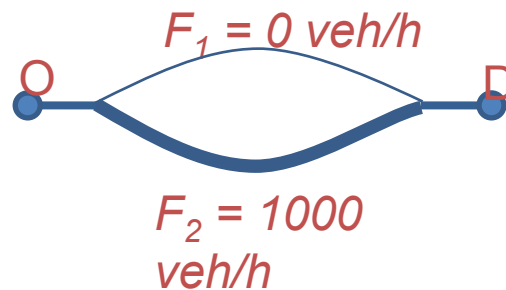


Example

New scenario: $t_2 = 15 \text{ min} \rightarrow 8 \text{ min}$



Deterministic



$$V_1 = \alpha \cdot t_1$$

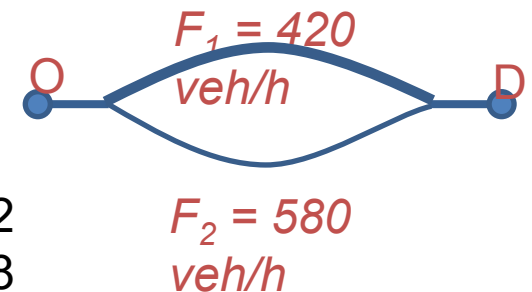
$$V_2 = \alpha \cdot t_2$$

$$\alpha = -5.90$$

$$\text{Prob}[1] = 0.42$$

$$\text{Prob}[2] = 0.58$$

Probabilistic



DETERMINISTIC MODELS

ADVANTAGES

- computational

DISADVANTAGES

- lack of realism
- path flow map is a multi-valued application

PROBABILISTIC MODELS

ADVANTAGES

- more realistic
- wider modelling flexibility
- path flow map is a continuous function

DISADVANTAGES

- computational

Random utility model

$$U_i = V_i(\text{attributes of } i; \text{parameters}) + \varepsilon_i$$

What is in ε ?

Analysts' imperfect knowledge:

- Unobserved attributes
- Unobserved taste variations
- Measurement errors
- Use of proxy variables

$$\begin{aligned} U(\text{bus}) &= \beta_1 \times (\text{walk_timebus}) + \beta_2 \times (\text{in_vehicle_timebus}) \\ &\quad + \beta_3 \times (\text{farebus} / \text{income}) + \varepsilon_{\text{bus}} \\ &= V(\text{bus}) + \varepsilon_{\text{bus}} \end{aligned}$$

- Attributes: describing the alternative
 - Generic vs. Specific
 - Examples: bus fare vs. travel time
 - Quantitative vs. Qualitative
 - Examples: reliability vs. comfort
 - Perception
 - Data availability

- Characteristics: describing the decision-maker
 - Socio-economic variables
 - Examples: income, gender, education

Random term (ε)

- Distribution of *epsilons* (ε)
 - Uniform $\rightarrow$ Linear
 - ...
 - Normal $\rightarrow$ Probit
 - Extreme Value or Gumbel $\rightarrow$ Logit

- Variance/covariance structure
 - Correlation between alternatives

Decision rule: Utility maximization

Decision maker n selects the alternative i with the highest utility U_{in} in the choice set C_n .

$$U_{in} = V_{in} + \varepsilon_{in}$$

V_{in} : Systematic utility expressed as a function of k observable variables, e.g. a linear-in-parameter specification. V_{in} is the deterministic part of the utility that the researcher can capture with observable attributes and their weights

ε_{in} : Random utility component

- Multinomial choice probability:

$$\begin{aligned}P(i|C_n) &= P(U_{in} \geq U_{jn}, \forall j \in C_n) \\P(i|C_n) &= P(U_{in} - U_{jn} \geq 0, \forall j \in C_n) \\P(i|C_n) &= P(U_{in} = \max_j U_{jn}, \forall j \in C_n)\end{aligned}$$

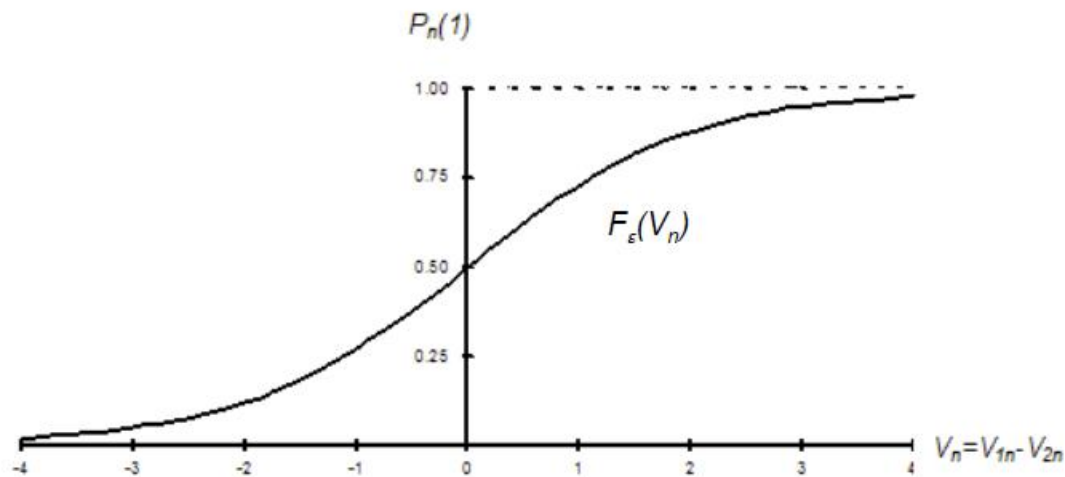
- Binary choice – Special case

$$\begin{aligned}P_n(1) &= P(U_{1n} \geq U_{2n}) \\P_n(1) &= P(U_{1n} - U_{2n} \geq 0)\end{aligned}$$

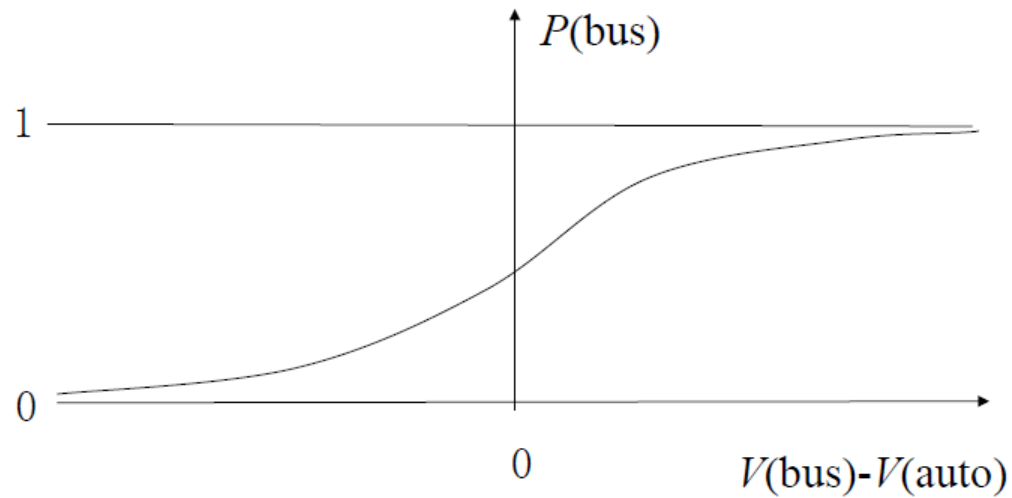
Binary choice

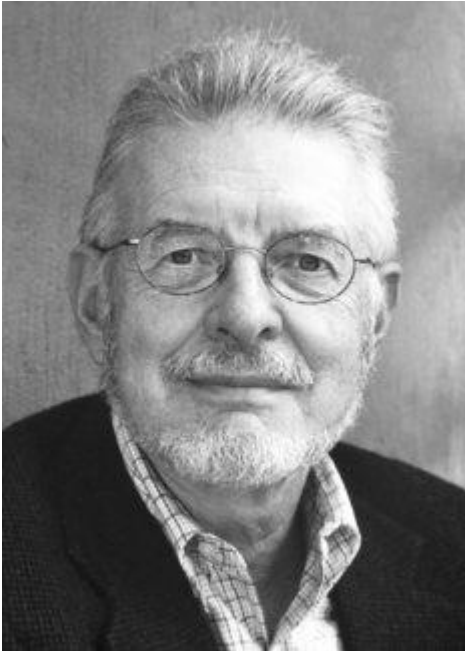
- Choice set $C_n = \{1,2\} \forall n$

$$\begin{aligned} P_n(1) &= P(1|C_n) = P(U_{1n} \geq U_{2n}) \\ &= P(V_{1n} + \varepsilon_{1n} \geq V_{2n} + \varepsilon_{2n}) \\ &= P(V_{1n} - V_{2n} \geq \varepsilon_{2n} - \varepsilon_{1n}) \\ &= P(V_{1n} - V_{2n} \geq \varepsilon_n) \\ &= P(V_n \geq \varepsilon_n) = F_\varepsilon(V_n) \end{aligned}$$



Probabilistic choice





Daniel McFadden

- 2000 Nobel Prize in Economic sciences
- His research on choice behaviour and the problem of linking economic theory and measurement.

Why the Gumbel distribution became popular (and the derivation of the family of models gave a Nobel prize to Daniel McFadden)?

- Gumbel distribution for the error terms is nearly the same as a normal distribution, but the choice probability is closed-form
- the difference between two error terms that are Gumbel distributed follows a logistic distribution and provides the closed-form formula

- The analytical solution of the integral yields the closed-form choice probability of the logit model

$$P_{ni} = \frac{\exp(V_{ni})}{\exp(V_{n1}) + \exp(V_{n2}) + \cdots + \exp(V_{nj})}$$

- The binary logit formula;

$$P_{n1} = \frac{\exp(V_{n1})}{\exp(V_{n1}) + \exp(V_{n2})} = \frac{1}{1 + \exp(V_{n2} - V_{n1})}$$

- V_{ni} involve constants that need to be calibrated
- Maximum Likelihood estimation
- Stated Preference or Revealed Preference data

Independence from irrelevant alternatives (IIA)

Probability of option i in the choice set C_n

$$P(i|C_n) = \frac{\exp(V_{in})}{\sum_{j \in C_n} \exp(V_{jn})}$$

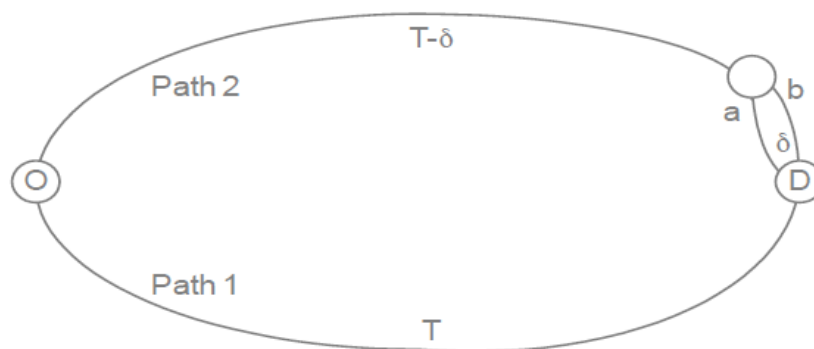
Odds ratio;

$$\frac{P(i|C_{1n})}{P(j|C_{1n})} = \frac{P(i|C_{2n})}{P(j|C_{2n})}$$

$$i, j \in C_{1n}, i, j \in C_{2n}, C_{1n} \subseteq C_n \text{ and } C_{2n} \subseteq C_n$$

*If A is preferred to B out of the choice set $\{A, B\}$, introducing a third option X , expanding the choice set to $\{A, B, X\}$, must not make B preferable to A . **Guarantees rational behaviour.***

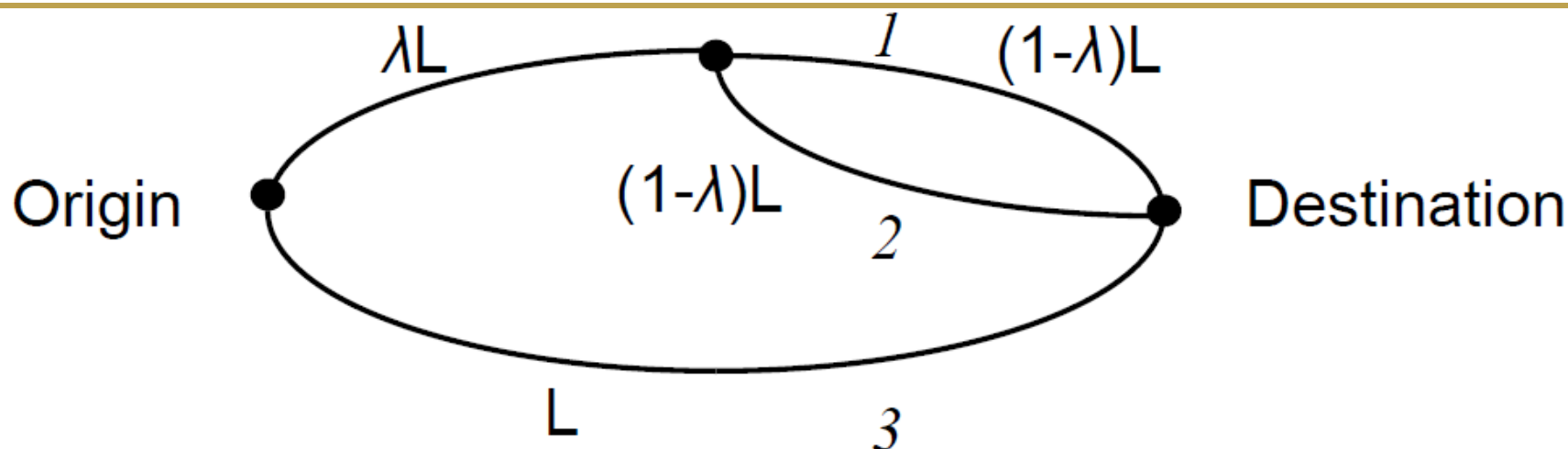
- Route choice example



$$P(1 | \{1, 2a, 2b\}) = P(2a | \{1, 2a, 2b\}) = P(2b | \{1, 2a, 2b\}) = \frac{e^{\mu T}}{\sum_{j \in \{1, 2a, 2b\}} e^{\mu T}} = \frac{1}{3}$$

- Paths may have overlapping sections
- Alternative paths may be correlated due to overlapping portions

Overlapping problem



- Introduces correlation - **Violates IIA** - Logit is unsuitable
- Traditional example with three paths, two overlapping
 - As overlap approaches 100 percent ($\lambda \rightarrow 1$), expect close to 25/25/50 shares
 - As overlap approaches 0 percent ($\lambda \rightarrow 0$), expect close to 33/33/33 shares
- Logit predicts 33/33/33 shares for any value of λ

- A variation of the logit model to address the commonality problem

$$P(i|C_n) = \frac{\exp(\mu V_{in} - CF_i)}{\sum_{j \in C_n} \exp(\mu V_{jn} - CF_j)}$$

- directly proportional to the degree of overlapping of path k with other alternative paths. Thus, highly overlapped paths have a larger CF factor and therefore smaller utility with respect to similar paths. CF_k is calculated as follows:

$$CF_i = \beta \cdot \ln \sum_{j \in C_n} \left(\frac{L_{ij}}{L_i^{1/2} L_j^{1/2}} \right)^\gamma$$

- L_{ij} is the cost of links common to paths i and j ,
- L_i and L_j are the cost of paths i and j respectively (expressed in hours if cost is travel time)
- Depending on the two factor parameters β and γ , a greater or lesser weighting is given to the 'commonality factor'.
- Larger values of β means that the overlapping factor has greater importance with respect to the utility V_i .
- The factor γ is a positive parameter, usually taken in the range $[0, 2]$, whose influence is smaller than β and which has the opposite effect on the choice.

- **Adaptive path choice**

- Apply path choice model repeatedly with certain frequency
- The more frequent, the closer to equilibrium
- The decision is based on instantaneous traffic conditions

- **Utility** usually consists of

- Travel time
- Link attractiveness
 - Some roads are preferred more than others because of a variety of reasons. This function allows the modeller to calibrate the simulation.

Next

This week

- Tutorial - Python